

Project: Superconducting qubit readout: an open quantum system perspective

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Introduction

The project aims to investigate the readout process of superconducting qubits in the presence of structured bosonic and spin environments. A full understanding of this process is of paramount relevance for the development of quantum computing devices.

It is well known that one of the major problems hindering the development of quantum computers is the presence of decoherence, energy loss, and, more generally, errors. All of these processes ensue from the interaction of the information support units — the qubits — with a large number of degrees of freedom, constituting, in the language of open quantum system theory, the environment. Understanding the effects of this interaction on the qubit is therefore a necessary step toward reducing these errors, and is currently the object of substantial ongoing investigation.

Superconducting qubit readout is, in particular, an instance of a measurement process. Measurements are, by definition, irreversible operations that in general destroy coherence and entanglement and induce energy loss. In quantum computing platforms, the readout of a qubit — i.e., the measurement of a selected observable — is typically performed by switching on an interaction between the qubit and a resonator, modelled as a harmonic mode of an electromagnetic field. The resonator is in turn coupled to a transmission line (TL), which carries the information about the qubit state to the actual measurement apparatus (e.g., a homodyne detection setup).

In the dispersive regime used for readout, the qubit and resonator are strongly detuned ($g \ll |\Delta|$), so there is no resonant exchange of real excitations between them — unlike, for instance, resonator-mediated gates, which operate in the resonant or near-resonant regime precisely to enable such exchange. Instead, a Schrieffer-Wolff transformation of the Jaynes-Cummings Hamiltonian yields an effective dispersive coupling term $\chi \hat{\sigma}_z$: the qubit shifts the resonator frequency by $\pm\chi$ depending on whether it occupies $|0\rangle$ or $|1\rangle$, while the resonator induces a small ac-Stark shift of the qubit frequency in return. Driving the resonator and detecting the resulting qubit-state-dependent phase and amplitude of the outgoing field — via homodyne/heterodyne measurement of the I and Q quadratures — therefore constitutes, ideally, a quantum non-demolition (QND) measurement of σ_z : since the dispersive coupling commutes with the qubit's own Hamiltonian, it does not change qubit level populations. The backaction that is accounted for in this idealized picture is not on the populations, but on the coherences: the act of scattering photons off the dispersively shifted resonator dephases superpositions of $|0\rangle$ and $|1\rangle$, since photon-number (shot-noise) fluctuations in the resonator translate into random phase kicks on the qubit. This designed-for measurement dephasing is not itself a problem — it is the expected price of extracting σ_z information at all.

What *is* a problem is that the qubit-resonator interaction can open additional interaction channels — energy pathways — to other degrees of freedom that would otherwise be inert (i.e., invisible to the qubit) in the absence of the coupling to the resonator. Among these degrees of freedom, spurious two-level

systems (TLS) in the amorphous solid embedding the qubit can play a significant role. If the interaction with these spins — enabled by activation of the qubit-resonator coupling — has characteristic timescales comparable to, or shorter than, the measurement timescale, the qubit state may be altered while it is being measured, leading to measurement errors. Recent work has begun to quantify this effect: Dai et al., Phys. Rev. X 16, 011011 (2026), "Characterization of drive-induced unwanted state transitions in superconducting circuits," and Huang et al., arXiv:2603.13857 (2026), "Readout-induced degradation of transmon lifetimes: interplay of TLSs and qubit spectral reshaping," both discuss the effects of qubit-TLS bath interactions on the qubit lifetime (T_1) and spectrum.

Outline of the Project

The aims of the project are:

1. Provide an understanding of circuit QED principles.
2. Verify, using numerically exact techniques, the quality of the approximations typically made when modelling the readout process.
3. Investigate the effects of a spurious TLS on the transmon-mode-TL dynamics and on the measurement outcomes.

To pursue these aims, we consider a sequence of scenarios, each addressing a different "cut" of the TLS-transmon-mode-TL complex, and in each we assess the deviations induced by commonly adopted approximations from the exact dynamics.

Scenario 0: Qubit–Resonator–TL

The qubit-mode interaction is modelled following Blais et al., Rev. Mod. Phys. 93, 025005 (2021), Section III.

- **0.1** — The transmon is coupled to a single harmonic mode via the full Hamiltonian interaction. This model illustrates the changes induced in the qubit energy levels by the presence of the resonator.
- **0.2** — The transmon is treated as a two-level system, and its reduced dynamics is described by a Lindblad master equation with spin-dependent rates.
- **0.3** — The transmon is coupled to the harmonic mode, whose free dynamics is in turn described by a Lindblad master equation representing its effective interaction with the TL, modelled as an unstructured (delta-correlated) bath. This effective Lindblad treatment is an approximation valid only under suitable conditions — among them, weak coupling between the resonator and the transmission line.
- **0.4** — The harmonic mode is coupled to the TL, this time modelled explicitly as a bosonic environment with a structured (Ohmic, Purcell-filtered-readout) spectral density.

Aims: Establishing whether model 0.3 shows significant deviations from the exact dynamics constitutes the first milestone of the project, and will determine how the readout mode is modelled in the remainder of the work. This part of the project has been already partially covered in Riva _et al._arXiv:2604.11722v1.

Methods: Models 0.1–0.3 can be solved either analytically or with basic numerical methods (integration of coupled differential equations). Model 0.4 requires non-perturbative, numerically exact methods (TEDOPA) to determine the qubit/resonator dynamics.

Scenario 1: Qubit + Spurious TLS

Here the qubit interacts with a single TLS.

- **1.1** — The TLS interacts only with the qubit. Analysis of the available transitions and of the effects induced on the qubit dynamics by the spurious TLS.
- **1.2** — The TLS additionally interacts with a bosonic bath with a super-Ohmic ($s = 3$) spectral density. We investigate how the dynamics found in 1.1 is modified by this secondary bath, which models the vibrational spectrum of the amorphous substrate surrounding the TLS.

Aims: Model 1.1 serves as an introduction to the dynamics of coupled spin systems; model 1.2 provides a building block for the simulation of the full system.

Methods: Model 1.1 is amenable to analytic and basic numerical treatment. Model 1.2 requires a numerically exact TN method, as in Scenario 0.

Scenario 2: Qubit + Resonator–TL + Spurious TLS

This is the complete scenario, built from the results and techniques developed in Scenarios 0 and 1.

The specific models investigated here depend on the outcome of the earlier scenarios. If Scenario 0.3 is confirmed as an accurate model for the qubit-resonator evolution, the complete model reduces to a transmon coupled to a damped-driven mode (driven pseudomode) and to a TLS interacting with a secondary super-Ohmic bath. If not, the full system instead comprises the transmon coupled to a mode and to a TLS, each interacting with its own secondary bath.

In this part of the project we will moreover introduce the driving field, which is used, via input-output formalism, to actually determine the state of the measured qubit.

The Project Within QLumen

QLumen's mission is to give early-master students a taste of what doing research means. This project is ambitious — arguably too extended for QLumen's 8-month time span. However, it is highly modular and has clear intermediate milestones. Participants will not be required to complete the whole project; what remains will be natural material for further work outside QLumen, e.g., a master's thesis.

Literature Roadmap by Thematic Block

(a) Circuit QED and Dispersive Readout

- Blais, Grimsmo, Girvin, and Wallraff, "Circuit Quantum Electrodynamics," Rev. Mod. Phys. 93, 025005 (2021) — covers the Jaynes–Cummings model, its dispersive limit, light-matter coupling regimes, and dispersive readout. Starting point for the qubit-resonator Hamiltonian and input-output formalism.
- Gardiner and Collett, Phys. Rev. A 31, 3761 (1985) — foundational reference for the input-output formalism invoked throughout the project.
- Gambetta, Blais, Boissonneault, Houck, Schuster, Girvin, Phys. Rev. A 77, 012112 (2008), "Photon-dependent qubit dephasing and relaxation rates" — theory of readout-induced dephasing/relaxation in the dispersive regime.
- Koch et al., Phys. Rev. A 76, 042319 (2007) — the transmon as a physical object.
- Wallraff et al., PRL 95, 060501 (2005); Walter et al., Phys. Rev. Applied 7, 054020 (2017) — experimental aspects of single-shot readout.

- Riva _et al._arXiv:2604.11722v1 - investigation of the qubit-resonator dynamics by means of numerically exact techniques.

(b) Microscopic Physics of TLSs

- Müller, Cole, and Lisenfeld, "Towards Understanding Two-Level-Systems in Amorphous Solids – Insights from Quantum Devices" — reference review on TLSs in amorphous solids applied to superconducting circuits.
- Phillips, J. Low Temp. Phys. 7, 351 (1972) — the foundational standard tunneling model.
- Martinis and Megrant, "Review of Decoherence and Materials Physics for Superconducting Qubits" (2014) — device-engineering perspective; identifies TLS defects in amorphous dielectrics as a primary decoherence mechanism, alongside non-equilibrium quasiparticles.
- Faoro and Ioffe, Phys. Rev. B 91, 014201 (2015), "Interacting Tunneling Model..." — on TLS-TLS interaction and the TLS bath as a noise source.
- Lisenfeld et al., Sci. Rep. 6, 23786 (2016); Nat. Commun. 6, 6182 (2015) — recent experimental phenomenology (spectroscopy of individual TLSs, resolved qubit-TLS coupling).
- W. Dai, S. Hazra, D. K. Weiss, P. D. Kurilovich, T. Connolly, H. K. Babla, S. Singh, V. R. Joshi, A. Z. Ding, P. D. Parakh, J. Venkatraman, X. Xiao, L. Frunzio, and M. H. Devoret, "Characterization of drive-induced unwanted state transitions in superconducting circuits," Phys. Rev. X 16, 011011 (2026) - addressing spectroscopic investigation.
- Z. Huang, J. S. Hung, M. Raha, M. Chou, H. Levine, A. Retzker, C. T. Hann, D. Hover, F. G. S. L. Brandão, A. A. Clerk, A. Haim, and O. Painter, "Readout-induced degradation of transmon lifetimes: interplay of TLSs and qubit spectral reshaping," arXiv:2603.13857 (2026)- impact on lifetime of qubits.

(c) Tensor-Network Methods for Open Systems (the Methodological Core)

This is the most valuable part of the roadmap, as it follows a clear line of development:

1. **Chain-mapping / TEDOPA** — transforming a continuous bosonic bath (star geometry) into a semi-infinite chain with nearest-neighbor couplings, tractable via DMRG/MPS:
 - Prior, Chin, Huelga, and Plenio, "Efficient Simulation of Strong System-Environment Interactions," PRL 105, 050404 (2010).
 - Chin, Rivas, Huelga, and Plenio, "Exact Mapping Between System-Reservoir Quantum Models and Semi-Infinite Discrete Chains Using Orthogonal Polynomials," J. Math. Phys. 51, 092109 (2010) — mathematical foundation (orthogonal polynomials) for chain-mapping.
 - Tamascelli, Smirne, Lim, Huelga, and Plenio, "Efficient Simulation of Finite-Temperature Open Quantum Systems," PRL 123, 090402 (2019) — extends the method to finite temperature, where naive chain-mapping scales poorly.
 - Practical implementation: **MPSDynamics.jl** (Chin's group), designed specifically for finite-temperature non-Markovian dynamics via tensor networks.
2. While mastering more than one TN-based method is beyond the scope of this project, a basic understanding of the broader landscape will be useful for future work. This project restricts attention to a single spurious TLS; a proper characterization of the effects of multiple TLSs on the measurement procedure is, however, a natural extension — for which other TN structures, such as tensor trees and their associated evolution algorithms, would become necessary.